

Methodological Concerns Regarding the Partially Randomized Design and Statistical Inference in Preoperative TEAS for Breast Cancer Patients: A Letter to the Editor [Response to Letter]

Can Ma^{1,2}, Hong Xiong³, Wenchao Fu¹, Shihua Lv¹, Lijuan Zhang^{1,2}, Dengming Zhao^{1,2}, Cong Hu¹, Xiuli Wang⁴, Wenzhi Li^{1,2}

¹The Key Laboratory of Anesthesiology and Intensive Care Research of Heilongjiang Province, Harbin, Heilongjiang, People's Republic of China;

²Department of Anesthesiology, Second Affiliated Hospital of Harbin Medical University, Harbin, Heilongjiang, People's Republic of China; ³The Third People's Hospital of Chengdu, Chengdu, Sichuan, People's Republic of China; ⁴Department of Anesthesiology, The Third Hospital of Hebei Medical University, Shijiazhuang, Hebei, People's Republic of China

Correspondence: Wenzhi Li, Department of Anesthesiology, Second Affiliated Hospital of Harbin Medical University, No. 246 Xuefu Road, Nangang District, Harbin, Heilongjiang, People's Republic of China, Tel +86 18045159998, Email wenzhili9@126.com

Dear editor

We sincerely thank Dr Wei¹ for the careful, rigorous, and constructive comments on our article.² The letter raises several genuinely important methodological questions, and we are grateful for the depth of scholarship with which they were articulated. We note that some of these concerns echo issues raised by the reviewers during peer review; the revisions undertaken at that stage — retitling the study as a partially randomized controlled trial, explicitly framing all comparisons involving Group C as observational, re-analyzing the longitudinal outcomes using two-way repeated-measures ANOVA with Bonferroni-corrected pairwise tests, expanding the reporting of the regression analyses, and softening the causal and mechanistic language — were intended to address them. Having studied Dr Wei's letter carefully, we accept that further clarification is warranted, and we respond to each comment below in a spirit of full acceptance of the underlying methodological principles.

Comment 1: First, the partially randomized design introduces fundamental challenges for causal inference. The observed “comparability” between Group T and Group C may merely reflect baseline differences rather than a genuine treatment effect. Regression to the mean cannot be excluded.

Response: We are grateful for this clear and important articulation. The partially randomized, stratified nature of the design was discussed extensively with the reviewers before publication, and the published article explicitly frames all comparisons involving Group C as observational (Abstract, Group Allocation and Randomization, Results, and the first stated Limitation). We fully agree with Dr Wei that comparisons between Group T and Group C cannot support confirmatory conclusions. The sentence “preoperative TEAS effectively normalized postoperative pain perception” was intended only to describe that the absolute NRS scores in Group T approximated those of Group C at the corresponding time points; we accept that this phrasing risks implying causal equivalence and that the regression-based associations should likewise be regarded as hypothesis-generating rather than confirmatory. We will convey these clarifications to the Editor in an appropriate form and, should the Editor consider it helpful, we are willing to append a formal clarification to the published record. Within the randomized stratum, Groups S and T were comparable at baseline (PSQI, AIS, and SAS; all $p > 0.05$), and the significantly lower postoperative NRS scores in Group T versus Group S support the effectiveness of the intervention; we are glad that Dr Wei's analysis likewise identifies this randomized comparison as the appropriate basis for the efficacy conclusion.

Comment 2: Second, the absence of adjustment for surgical procedure type in regression analyses represents a major confounding concern. The authors should have performed multivariable regression or ANCOVA adjusting for at least surgical procedure type, baseline anxiety, and intraoperative opioid consumption to determine whether PSQI remains an independent predictor.

Response: We fully accept this point. As Dr Wei rightly notes, acute postoperative pain is influenced by many factors, and adjustment for surgical procedure type, baseline anxiety, and intraoperative opioid consumption would be required to establish PSQI as an independent predictor. Our study was not designed to identify independent predictive factors; we sought only to examine whether preoperative PSQI is associated with acute postoperative pain and sleep quality, and we deliberately kept the regression models parsimonious given the modest size of the combined non-TEAS cohort ($n = 42$). Although the distribution of surgical procedures and surgical duration did not differ significantly among the groups (Table 1), we agree that statistical balance does not eliminate residual confounding. We are grateful for this suggestion and will incorporate multivariable regression or ANCOVA with appropriate adjustment — including surgical procedure type, baseline anxiety, and perioperative analgesic consumption — in our future studies, in order to determine whether PSQI is an independent predictor of postoperative outcomes.

Comment 3: Third, the absence of a sham-TEAS control group limits the ability to exclude placebo effects. We encourage the authors to frame their findings as preliminary evidence requiring confirmation in a sham-controlled, fully randomized trial.

Response: We fully agree, and we thank Dr Wei for emphasizing this point. The absence of a sham-TEAS control was acknowledged in the published Limitations, but we accept that it is more than a minor methodological nuance: it fundamentally affects the interpretation of the subjective outcomes (NRS and AIS). While we note that the intraoperative endpoints (sdSBP, cvSBP, and ephedrine consumption) were recorded under general anesthesia and are therefore less susceptible to expectation effects, we accept that this does not address the fundamental point regarding the patient-reported outcomes. Accordingly, we will explicitly define the conclusions of this study as preliminary evidence, and we are committed to confirming them in a sham-controlled, fully randomized clinical trial, which our group is now planning.

Comment 4: Fourth, the designation of NRS as a co-primary outcome without formal sample size calculation warrants methodological scrutiny. The authors should consider reclassifying NRS as a secondary or exploratory outcome.

Response: We thank Dr Wei for this valuable recommendation, and we accept it. In retrospect, designating NRS as a co-primary outcome without a formal a priori power calculation — and relying on post-hoc inspection to justify the sample size — was inappropriate and could increase the risk of false-positive findings. With the Editor's agreement, we will adjust the outcome hierarchy by issuing a clarification that designates postoperative sleep quality (AIS) — the outcome on which the sample size calculation was based — as the single primary outcome, and reclassifies postoperative NRS pain intensity as a secondary, exploratory outcome, interpreting the pain findings accordingly. For completeness, we note that the longitudinal analyses were performed within a repeated-measures framework with Bonferroni-corrected pairwise comparisons; nevertheless, we agree that appropriate outcome designation, rather than analytic correction alone, is the proper remedy, and we are grateful to Dr Wei for guiding us toward it.

In summary, we are sincerely grateful to Dr Wei for the rigor, fairness, and constructiveness of this critique. Our original aspiration in conducting this study was to bring techniques derived from traditional Chinese medicine, such as TEAS, into evidence-based perioperative practice; at the same time, clinical research is subject to many influences, and we accept without reservation the limitations that have been identified. The issues raised in this letter — the primacy of the randomized comparison, the need for adjusted analyses, sham controls, and disciplined outcome designation — will directly inform the design of our planned sham-controlled, fully randomized, multicenter trial. We warmly welcome any further comments from Dr Wei and from the readership, and we thank the Editor for the opportunity to respond.

Disclosure

The authors report no conflicts of interest in this communication.

References

1. Wei M, Tong J, Pang Y, Zhou X, Qiu Y, Wang Z. Methodological concerns regarding the partially randomized design and statistical inference in preoperative TEAS for breast cancer patients: a letter to the editor. *J Pain Res.* 2026;19. doi:10.2147/JPR.S617864
2. Ma C, Xiong H, Fu W, et al. Preoperative transcutaneous electrical acupoint stimulation alleviates postoperative pain and improves sleep quality in breast cancer patients with preoperative sleep disturbance: a partially randomized controlled trial. *J Pain Res.* 2026;19:1–14. doi:10.2147/JPR.S619125

Dove Medical Press encourages responsible, free and frank academic debate. The content of the Journal of Pain Research 'letters to the editor' section does not necessarily represent the views of Dove Medical Press, its officers, agents, employees, related entities or the Journal of Pain Research editors. While all reasonable steps have been taken to confirm the content of each letter, Dove Medical Press accepts no liability in respect of the content of any letter, nor is it responsible for the content and accuracy of any letter to the editor.

Journal of Pain Research

Publish your work in this journal

The Journal of Pain Research is an international, peer reviewed, open access, online journal that welcomes laboratory and clinical findings in the fields of pain research and the prevention and management of pain. Original research, reviews, symposium reports, hypothesis formation and commentaries are all considered for publication. The manuscript management system is completely online and includes a very quick and fair peer-review system, which is all easy to use. Visit <http://www.dovepress.com/testimonials.php> to read real quotes from published authors.

Submit your manuscript here: <https://www.dovepress.com/journal-of-pain-research-journal>

Dovepress
Taylor & Francis Group